
REHABILITAR: ACTIVITY-AWARE AUGMENTED REALITY INSTRUCTIONS FOR REHABILITATION

MSC PROJECT WORK - 10 ECTS

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PROJECT REPORT

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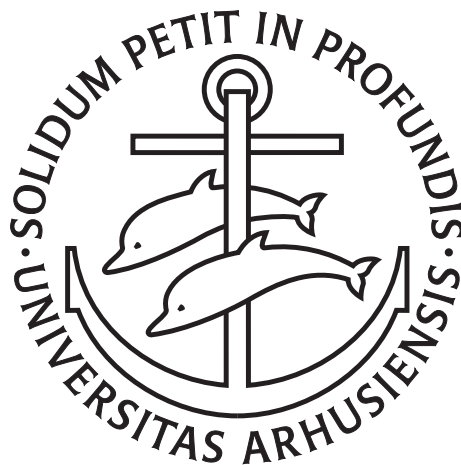


AARHUS
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DEPARTMENT OF COMPUTER SCIENCE

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Department of Computer Science
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ABSTRACT

Shoulder rehabilitation requires consistent exercise, but traditional physiotherapy often suffers from low adherence due to lack of engagement and delayed feedback. RehabilitAR is an augmented reality (AR) application for Meta Quest 3, utilizing activity-based tracking-free guidance to improve shoulder rehabilitation. Using passthrough AR, inverse kinematics (IK) for pose estimation, and multi-layered visual aids (virtual mirror, 3D avatar, trainer videos), it provides real-time feedback and gamified progression for exercises like front raises and lateral raises. A user study with 10 participants evaluated usability via the System Usability Scale (SUS, mean 81, "Excellent") and a 13-question custom questionnaire, highlighting strengths in movement clarity and areas for improvement in rep tracking. RehabilitAR contributes an accessible and cost-effective AR solution, advancing home-based rehabilitation.

1

INTRODUCTION

Shoulder injuries, such as rotator cuff tendinopathy, affect millions of people worldwide, necessitating effective rehabilitation to restore function and reduce pain [13]. Traditional physiotherapy faces challenges in patient adherence due to repetitive exercises and lack of immediate feedback [4]. Augmented reality (AR) and virtual reality (VR) offer immersive solutions to enhance motivation and provide real-time feedback, as demonstrated in movement training [1] and sports coaching [18]. Moreover, many AR/VR systems rely on external tracking devices or wearable sensors [6], limiting, in this way, the use of applications by users who do not have or do not want to buy hardware for only one application.

RehabilitAR addresses these gaps by developing an activity-aware AR application for shoulder rehabilitation, running on Meta Quest 3 without external trackers. Unlike systems using Vive trackers [6], RehabilitAR leverages the Quest's built-in tracking and passthrough AR to deliver a cost-effective, accessible solution, that enhances engagement and performance [10]. It integrates inverse kinematics (IK) for precise pose estimation, a virtual mirror for real-time posture feedback, 3D avatar animations, and trainer videos to guide users through clinically relevant exercises (front raises, lateral raises, arm circles, front-to-side raises). The technical novelties include tracker-free pose estimation, multi-layered guidance, and seamless passthrough AR integration, enhancing engagement and accuracy in a home-based setting.

This work contributes: (1) a novel AR prototype for shoulder rehabilitation, (2) a tracker-free, activity-aware feedback system, and (3) a usability evaluation through SUS and a custom questionnaire, evaluating effectiveness and user satisfaction. A user study with 10 participants validated the app's usability, achieving an "Excellent" SUS score of 81 [2]. This paper details the methodology, user study, results, and implications, aiming to advance AR-based rehabilitation solutions.

2

RELATED WORK

Augmented and virtual reality (AR/VR) technologies have significantly impacted rehabilitation and sports training [16, 17]. Several systems have explored these domains: for example, Wen et al. [18] developed an AR coaching solution using volumetric motion annotations, relying on external tracking setups; Farley et al. [9] used VR to promote back-friendly lifting techniques; and Kim, Hong, and Kim [11] proposed an AR-based remote coaching system with real-time feedback capabilities, focusing on general sports coaching. In parallel, XR serious games and exergames have gained popularity in healthcare settings due to their motivational benefits in rehabilitation contexts [8].

Despite these advances, existing solutions often require additional hardware (e.g., external sensors or trackers), target general or lower-body training, or lack specialization in clinically guided rehabilitation protocols. Moreover, few systems focus on shoulder-specific recovery, an area that demands careful attention to form, movement phases, and muscle activation patterns.

RehabilitAR addresses these challenges through a tracker-free AR design that simplifies the setup and enhances usability. It provides shoulder-targeted rehabilitation using evidence-based exercises—such as front raises, lateral raises, arm circles, and front-to-side raises—proven effective for deltoid strengthening and early-to-mid stage recovery [5, 7, 14, 15]. The system uses inverse kinematics (IK)-based tracking to deliver real-time feedback, ensuring correct form without requiring external equipment. Through this combination of clinically relevant exercises, hardware-free deployment, and layered instructional guidance, RehabilitAR contributes a practical and focused solution for shoulder rehabilitation.

3

METHODOLOGY

3.1 RESEARCH GOAL

The goal is to develop and evaluate an AR-based application for shoulder rehabilitation that enhances usability, engagement, and exercise accuracy without external trackers. RehabilitAR addresses the following research question: Can a tracker-free AR app achieve high usability for shoulder rehabilitation? This question tackle low adherence in traditional physiotherapy [4] and the need for accessible home-based solutions.

3.2 RESEARCH PROTOTYPE

RehabilitAR is a sophisticated augmented reality (AR) application developed in Unity 2022.3 for the Meta Quest 3, utilizing the Meta Quest 3 SDK and XR Interaction Toolkit to enable seamless controller-based interactions in passthrough mode, which blends virtual AR overlays with the real-world environment. Unlike systems relying on external hardware, such as Vive trackers [6], RehabilitAR leverages the Quest's built-in hand and controller tracking, ensuring a cost-effective and accessible solution that minimizes setup complexity for home-based rehabilitation. The passthrough AR mode was chosen to maintain users' environmental awareness, reducing potential discomfort associated with prolonged VR use. The prototype employs inverse kinematics (IK) through the custom 'UpperBodyIKDualConfig.cs' script, which maps controller inputs to a 3D avatar's arm movements, enabling real-time pose estimation. This IK system ensures precise tracking of shoulder exercises by validating arm positions against target angles, supporting the app's activity-aware feedback mechanism.

The prototype's user guidance features are designed to provide clear, multi-layered feedback, enhancing exercise accuracy and engagement. These include a virtual mirror, which reflects the user's IK-mapped avatar movements in real-time to facilitate posture correction; a 3D avatar with looping animations via Unity's Animator to demonstrate correct exercise form; and embedded trainer videos offering relatable, real-world references for technique.

Key design decisions were driven by the goal of creating an accessible, engaging, and clinically relevant rehabilitation tool. The tracker-free approach was selected to eliminate the need for costly external hardware. Passthrough AR was chosen over full VR to maintain situ-

ational awareness, enhancing user comfort and safety. The IK-based tracking system was implemented to balance accuracy with simplicity, leveraging the Quest's controllers to map arm movements effectively. The combination of multilayered guidance and visual aids was designed to accommodate diverse user needs. These decisions collectively ensure RehabilitAR is a user-friendly and effective tool for shoulder rehabilitation.

3.3 EXERCISE SELECTION

RehabilitAR includes front raises, lateral raises, arm circles, and front-to-side raises, selected for their efficacy in shoulder rehabilitation. Campos et al. [5] found front raises activate the anterior deltoid, while lateral raises target the middle deltoid. Arm circles promote mobility for early-phase rehabilitation [14], and front-to-side raises align with Phase III recovery [15].

3.4 USER GUIDANCE FEATURES

To guide users, RehabilitAR integrates three visual aids, as shown in Fig. 3.1:

- **Virtual Avatar Demonstration:** There is a 3D avatar that is continuously animating each exercise using Unity's Animator, showing the correct form. The animation was created manually based on videos from trainers/physiotherapists and is also synced with the Trainer Video in the application.
- **Trainer Video:** Embedded real-life video clips provide a relatable reference.
- **Virtual Mirror:** An IK-mapped avatar reflects user movements in real-time.

Pressing the Y button toggles a UI with textual instructions for each exercise (Fig. 3.2), while A skips exercises and B exits the app. Fig. 3.3 shows a user interacting with the system.

3.5 REPETITION COUNTING

Repetition counting is one of the most important features of this application. When a user does rehabilitation, they have to implement the exercises correctly. Thus, I needed a mechanism that tracks the repetitions so that the user could do each exercise correctly and be able to see that they were successful. Repetitions are tracked using inverse



Figure 3.1: Screenshot of RehabilitAR showing the virtual mirror, 3D avatar, and trainer video.

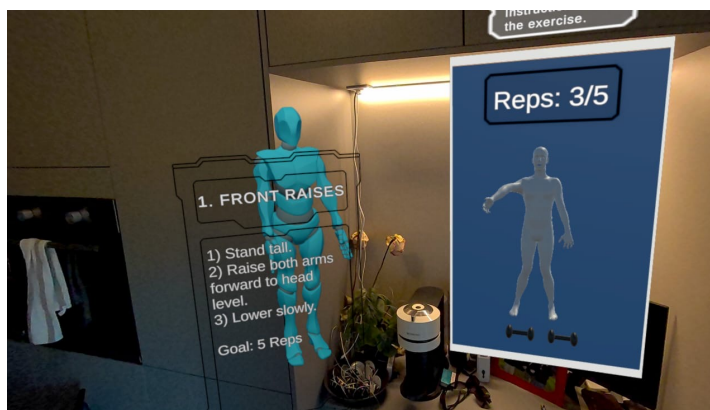


Figure 3.2: Instructions UI for current exercise.

kinematics (IK) in the ‘UpperBodyIKDualConfig.cs’ script, mapping Quest controller inputs to a virtual avatar’s arms. Arm positions (e.g., Front, Side) are validated by comparing quaternions against target angles within an ‘angleTolerance’ of 25 degrees, counting a rep when the arm completes a sequence (e.g., Down → Front → Side). Green/Red overlays indicate success/fast repetitions (e.g., exceeding ‘maxVelocity’). Adjustments to ‘angleTolerance’ (to 30 degrees instead of 25) for some exercises resolved repetition tracking issues that occurred during the implementation of the application.

3.6 CODE AVAILABILITY & DEMO

I provide a link to the GitHub repository associated with this project: [RehabilitAR](#). This repository serves as a comprehensive resource for readers interested in exploring the source code, accessing additional materials, or furthering their understanding of the concepts discussed in the project. By providing access to the repository, I aim to pro-



Figure 3.3: User performing shoulder exercises with RehabilitAR on the Meta Quest 3.

mote transparency, reproducibility, and collaboration in the realm of academic research and software development.

In the following link, you will find a video demonstrating a presentation of the app: [RehabilitAR Demo](#).

4

USER STUDY

4.1 GOALS AND RESEARCH QUESTIONS

The user study evaluated RehabilitAR's usability and guidance effectiveness, addressing the following question: Is the app usable for shoulder rehabilitation? These questions are critical as usability drives adoption, and effective guidance improves form [2].

4.2 STUDY DESIGN

Ten participants (aged 20–30, 6 male, 4 female, mixed AR experience, mixed rehabilitation experience) were recruited. Each performed the four exercises in a controlled setting using the Meta Quest 3 (Fig. 3.3), completing questionnaires: the paper-based SUS (10 questions, 1-5 Likert scale) and a 13-question custom questionnaire (1-5 Likert scale). Each of them completed the testing of the application in approximately 5-7 minutes before they answered the questionnaires. The SUS questionnaire provided standardized usability metrics [2], while the custom questionnaire assessed app-specific features. Data were analyzed using Python for means, medians and standard deviations.

4.3 CUSTOM QUESTIONNAIRE DESIGN

The custom questionnaire evaluated usability, feedback clarity, and exercise suitability through 13 questions (Table 4.1). I developed these questions thinking which features were most important for: (1) The usability of the application, (2) The rehabilitation goals of the application, (3) The AR part of the application. After that, I slightly modified them using feedback from my supervisor. Responses used a 1-5 Likert scale (1 = strongly disagree, 5 = strongly agree).

Table 4.1: Custom Questionnaire Questions

Q#	Question
Q1	The app correctly counted my reps when performing the exercises.
Q2	The red overlay clearly indicated if a move was too fast.
Q3	The app recognized my arm positions as expected.
Q4	The movements I needed to do for each exercise were clear.
Q5	The green overlay helped me understand that I completed a rep correctly.
Q6	I would prefer more guidance (e.g., visual cues, audio instructions).
Q7	It was comfortable using the VR controllers for these exercises.
Q8	The avatar's arm movements matched my real-world movements well.
Q9	The A button skip feature was easy to use and worked as expected.
Q10	The required pace of the exercises felt natural.
Q11	It was challenging to complete the required reps (e.g., 5 reps).
Q12	The app transitioned smoothly to the next scene after finishing reps or pressing A.
Q13	I would like additional features (e.g., progress tracking, different exercises).

5 | RESULTS

5.1 SYSTEM USABILITY SCALE

RehabilitAR achieved an average SUS score of 81 across 10 user evaluations, rated “Excellent” by Bangor, Kortum, and Miller [3], exceeding their threshold of 80.3 for high usability (Fig. 5.1). This score earns an A grade, places the app in the top 25% of systems (4th quartile), and is deemed “Acceptable,” surpassing the average of 68. The median of 83.75 indicates consistent satisfaction, though a standard deviation of 10.35 suggests variability due to differing AR familiarity.

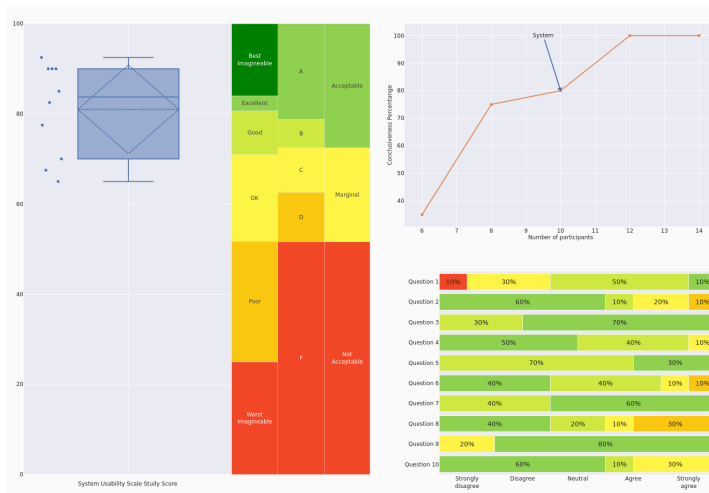


Figure 5.1: SUS results: Boxplot, scale, conclusiveness chart, and response distribution.

5.2 CUSTOM QUESTIONNAIRE

The custom questionnaire (1–5 Likert scale) revealed strong agreement across 13 questions (Fig. 5.2, 5.3). The highest scores were for the buttons features (Q9, mean 4.90, SD 0.32, median 5.00), clear exercise movements (Q4, mean 4.80, SD 0.42, median 5.00), and smooth scene transitions (Q12, mean 4.70, SD 0.48, median 5.00), with low standard deviations and high medians indicating strong user agreement. Rep counting (Q1, mean 4.10, SD 0.74, median 4.00) and the red overlay for fast moves (Q2, mean 4.10, SD 0.74, median 4.00) scored well but showed more variability, suggesting inconsistent tracking for some users. Users desired more guidance (Q6, mean 3.30, SD 0.82, median

3.00), found reps moderately challenging (Q11, mean 3.10, SD 1.52, median 3.50), and had mixed interest in features like progress tracking (Q13, mean 3.90, SD 1.10, median 4.00), with higher standard deviations and slightly lower medians reflecting diverse fitness levels and preferences.

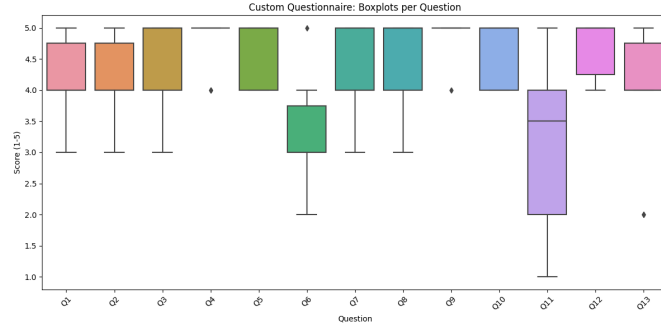


Figure 5.2: Boxplot visualization of custom questionnaire responses showing medians.

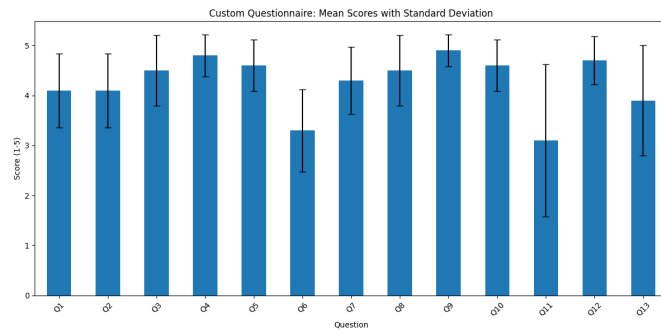


Figure 5.3: Mean scores with standard deviation for the custom questionnaire.

5.3 DISCUSSION

The SUS score of 81 suggests that RehabilitAR is an effective and highly usable AR application for training and rehabilitation. High score for Q4 highlights the effectiveness of visual aids (Fig. 3.1) and intuitive controls, likely due to the virtual mirror and avatar animations. Lower scores for Q1 and Q2 indicate tracking issues, with qualitative feedback suggesting ‘angleTolerance’ refinement. The high SD for Q11 may reflect varied fitness levels or rehabilitation experience, a limitation of the small sample ($n=10$). Compared to systems using external trackers [6], RehabilitAR’s tracker-free design runs on a standalone device but requires improved tracking accuracy.

6

CONCLUSION & FUTURE WORK

RehabilitAR delivers a user-friendly, tracker-free augmented reality (AR) application. Using pass-through AR, inverse kinematics (IK) for precise pose estimation, and multilayered guidance, including a virtual mirror, 3D avatar animations, and embedded trainer videos, RehabilitAR offers an immersive, cost-effective solution for home rehabilitation. The user study with 10 participants demonstrated exceptional usability, achieving an “Excellent” System Usability Scale (SUS) score of 81, placing it among the top 25% of evaluated systems [3]. The custom questionnaire underscored strengths in movement clarity and intuitive controls, while identifying areas for refinement, such as rep tracking accuracy, which informs future development.

However, limitations must be acknowledged: the small sample size (n=10) limits generalizability, the lack of clinical patient testing restricts insights into therapeutic efficacy, and tracking inaccuracies, highlight the constraints of controller-based IK in capturing complex movements. Potential adverse effects, such as discomfort from prolonged VR use, known as cybersickness, also warrant further investigation.

Future development will focus on multiple enhancements to broaden RehabilitAR’s impact. Refining tracking precision by increasing ‘angle-Tolerance’ to 35 degrees and incorporating audio cues, will improve user feedback. Integrating external sensors, such as wearable inertial units or depth cameras, could enhance motion capture accuracy [6]. Extending the system to support rehabilitation for additional body parts, such as lower-body or multi-joint exercises, will increase versatility. Adopting **Final IK** [12], an advanced inverse kinematics library, could improve avatar realism and pose estimation. Additionally, implementing progress tracking and conducting large-scale clinical trials with diverse patient populations will validate therapeutic outcomes. Through these advancements, RehabilitAR has the potential to transform rehabilitation practices, leveraging AR to deliver engaging, clinically relevant, and widely accessible solutions.

7

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BIBLIOGRAPHY

- [1] Fraser Anderson, Tovi Grossman, Justin Matejka, and George Fitzmaurice. "YouMove: enhancing movement training with an augmented reality mirror." In: *Proceedings of the 26th annual ACM symposium on User interface software and technology*. 2013, pp. 311–320.
- [2] Aaron Bangor, Philip T Kortum, and James T Miller. "An empirical evaluation of the system usability scale." In: *Intl. Journal of Human–Computer Interaction* 24.6 (2008), pp. 574–594.
- [3] Aaron Bangor, Philip Kortum, and James Miller. "Determining what individual SUS scores mean: Adding an adjective rating scale." In: *Journal of usability studies* 4.3 (2009), pp. 114–123.
- [4] Rona Campbell, M Evans, M Tucker, B Quilty, Paul Dieppe, and JL Donovan. "Why don't patients do their exercises? Understanding non-compliance with physiotherapy in patients with osteoarthritis of the knee." In: *Journal of Epidemiology & Community Health* 55.2 (2001), pp. 132–138.
- [5] Yuri AC Campos, Jeferson M Vianna, Miller P Guimarães, Jorge LD Oliveira, Claudio Hernández-Mosqueira, Sandro F da Silva, and Paulo H Marchetti. "Different shoulder exercises affect the activation of deltoid portions in resistance-trained individuals." In: *Journal of human kinetics* 75 (2020), p. 5.
- [6] Polona Caserman, Augusto Garcia-Agundez, Robert Konrad, Stefan Göbel, and Ralf Steinmetz. "Real-time body tracking in virtual reality using a Vive tracker." In: *Virtual Reality* 23 (2019), pp. 155–168.
- [7] Giuseppe Coratella, Gianpaolo Tornatore, Stefano Longo, Fabio Esposito, and Emiliano Cè. "An electromyographic analysis of lateral raise variations and frontal raise in competitive body-builders." In: *International journal of environmental research and public health* 17.17 (2020), p. 6015.
- [8] Robertas Damaševičius, Rytis Maskeliūnas, and Tomas Blažauskas. "Serious games and gamification in healthcare: a meta-review." In: *Information* 14.2 (2023), p. 105.
- [9] Andreas Farley, Xingyao Yu, André Tomalka, Tobias Siebert, and Michael Sedlmair. "LiftVR: A VR-Based Training System for Back-Friendly Lifting." In: *2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. IEEE. 2025, pp. 671–678.

- [10] Zixuan Guo, Hanxiao Deng, Hongyu Wang, Angel JY Tan, Wenge Xu, and Hai-Ning Liang. "Exploring the impact of passthrough on vr exergaming in public environments: A field study." In: *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE. 2024, pp. 229–238.
- [11] Youngsun Kim, Seokjun Hong, and Gerard J Kim. "Augmented reality based remote coaching system." In: *Proceedings of the 22nd ACM Conference on Virtual Reality Software and Technology*. 2016, pp. 311–312.
- [12] Pärtel Lang. *Inverse kinematics in dead and buried*. 2016.
- [13] Chris Littlewood, Jon Ashton, Ken Chance-Larsen, Stephen May, and Ben Sturrock. "Exercise for rotator cuff tendinopathy: a systematic review." In: *Physiotherapy* 98.2 (2012), pp. 101–109.
- [14] Joy L Long, Ramon A Ruberte Thiele, Jack G Skendzel, Jongeun Jeon, Richard E Hughes, Bruce S Miller, and James E Carpenter. "Activation of the shoulder musculature during pendulum exercises and light activities." In: *journal of orthopaedic & sports physical therapy* 40.4 (2010), pp. 230–237.
- [15] Peter D McCann, Mary E Wootten, Murali P Kadaba, and Louis U Bigliani. "A kinematic and electromyographic study of shoulder rehabilitation exercises." In: *Clinical Orthopaedics and Related Research (1976-2007)* 288 (1993), pp. 179–188.
- [16] Margarida F Pereira, Cosima Prahm, Jonas Kolbenschlag, Eva Oliveira, and Nuno F Rodrigues. "Application of AR and VR in hand rehabilitation: A systematic review." In: *Journal of biomedical informatics* 111 (2020), p. 103584.
- [17] Pooya Soltani and Antoine HP Morice. "Augmented reality tools for sports education and training." In: *Computers & Education* 155 (2020), p. 103923.
- [18] Jiqing Wen, Lauren Gold, Qianyu Ma, and Robert LiKamWa. "Augmented coach: Volumetric motion annotation and visualization for immersive sports coaching." In: *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. IEEE. 2024, pp. 137–146.